Міністерство освіти і науки України
Національної Університет “Львівська Політехніка”

Лабораторна робота
з дисципліни
“Об'єктно-орієнтоване програмування, частина 2”

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Лабораторна робота 9

Тема: Візуалізація даних за допомогою бібліотеки Seaborn

Мета: Навчитися працювати з бібліотекою Seaborn для візуалізації даних. Ознайомитися з основними типами графіків, які надає бібліотека, навчитися застосовувати їх для аналізу даних.

Завдання:

```
!pip install gdown

import gdown
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

sns.set(style="whitegrid")

file_id = "1TRTuKZyHJspxBuJTgJ_5JxbQX-v7kkJB"
url = f"https://drive.google.com/uc?id={file_id}"

gdown.download(url, "Job opportunities.csv", quiet=False)

class DataPreprocessor:

    def __init__(self, file_name):
        self.df = pd.read_csv(file_name)

    def create_average_salary(self):

        salary = self.df['Salary Range']
        salary = salary.str.replace('$', '', regex=False)
        salary = salary.str.replace('£', '', regex=False)
        salary = salary.str.replace(',', '', regex=False)
        salary = salary.str.strip()

        salary_split = salary.str.split('-', expand=True)

        self.df['Min Salary'] = salary_split[0].astype(float)
        self.df['Max Salary'] = salary_split[1].astype(float)

        self.df['Average Salary'] = (
            self.df['Min Salary'] + self.df['Max Salary']
        ) / 2

    def create_year(self):
```

```
        self.df['Date Posted'] = pd.to_datetime(self.df['Date Posted'])
        self.df['Year'] = self.df['Date Posted'].dt.year

    def get_data(self):

        self.create_average_salary()
        self.create_year()
        return self.df


class Visualizer:

    def __init__(self, df):
        self.df = df

    def barplot_salary_experience(self):

        plt.figure(figsize=(10,6))
        sns.barplot(
            x='Experience Level',
            y='Average Salary',
            data=self.df
        )
        plt.title('Average salary by experience')
        plt.show()

        print("Salary increases with experience")

    def boxplot_industry(self):

        plt.figure(figsize=(12,6))
        sns.boxplot(
            x='Industry',
            y='Average Salary',
            data=self.df
        )
        plt.xticks(rotation=45)
        plt.title('Salary by industry')
        plt.show()
```

```python
        print("Different industries have different salaries")

    def heatmap_jobs(self):

        table = pd.crosstab(
            self.df['Experience Level'],
            self.df['Industry']
        )

        plt.figure(figsize=(12,6))
        sns.heatmap(table, annot=True)
        plt.title('Jobs count')
        plt.show()

        print("Most jobs are Mid and Senior")

    def scatter_salary_year(self):

        plt.figure(figsize=(10,6))
        sns.scatterplot(
            x='Year',
            y='Average Salary',
            hue='Experience Level',
            data=self.df
        )
        plt.title('Salary vs year')
        plt.show()

        print("Salary grows with year")

    def pairplot_data(self):

        sns.pairplot(
            self.df[['Average Salary','Year','Experience Level']],
            hue='Experience Level'
        )
        plt.show()

        print("Relationships between variables")
```

```python
file_name = "Job opportunities.csv"

prep = DataPreprocessor(file_name)
df = prep.get_data()

print(df.head())

viz = Visualizer(df)

viz.barplot_salary_experience()
viz.boxplot_industry()
viz.heatmap_jobs()
viz.scatter_salary_year()
viz.pairplot_data()

print("Seaborn helps visualize and analyze data")
```

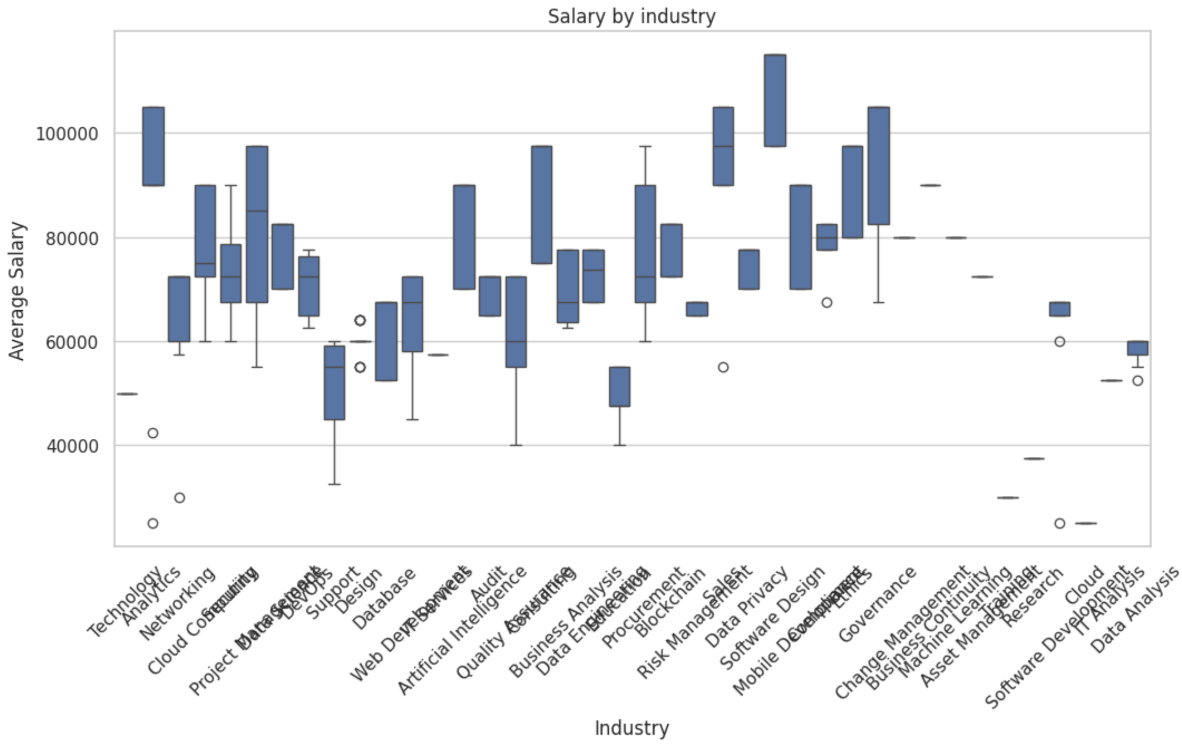
Результат:

```
Requirement already satisfied: gdown in /usr/local/lib/python3.12/dist-packages (5.2.1)
Requirement already satisfied: beautifulsoup4 in /usr/local/lib/python3.12/dist-packages (from gdown) (4.13.5)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from gdown) (3.25.2)
Requirement already satisfied: requests[socks] in /usr/local/lib/python3.12/dist-packages (from gdown) (2.32.4)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from gdown) (4.67.3)
Requirement already satisfied: soupsieve>1.2 in /usr/local/lib/python3.12/dist-packages (from beautifulsoup4->gdown) (2.8.3)
Requirement already satisfied: typing-extensions>=4.0.0 in /usr/local/lib/python3.12/dist-packages (from beautifulsoup4->gdown) (4.15.0)
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests[socks]->gdown) (3.4.6)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests[socks]->gdown) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests[socks]->gdown) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests[socks]->gdown) (2026.2.25)
Requirement already satisfied: PySocks!=1.5.7,>=1.5.6 in /usr/local/lib/python3.12/dist-packages (from requests[socks]->gdown) (1.7.1)
Downloading...
From: https://drive.google.com/uc?id=1TRTuKZyHJspxBuJTgJ_5JxbQX-v7kkJB
To: /content/Job opportunities.csv
100%|██████████| 56.4k/56.4k [00:00<00:00, 61.7MB/s]
                Job Title    Job Description Required Skills  \
0       Software Engineer   Develop software    Java, Python
1            Data Analyst       Analyze data      SQL, Excel
2        Network Engineer  Maintain networks      Cisco, WAN
3         Cloud Architect       Design cloud      AWS, Azure
4  Cybersecurity Analyst        Protect data   Cybersecurity

       Salary Range    Location             Company Experience Level  \
0  £40,000 - £60,000      London            ABC Tech      Entry-Level
1  £35,000 - £50,000  Manchester       XYZ Analytics           Junior
2  £45,000 - £70,000  Birmingham  Network Solutions        Mid-Level
3  £60,000 - £90,000   Edinburgh    Cloud Innovators           Senior
4  £50,000 - £80,000     Glasgow         SecureGuard        Mid-Level

          Industry   Job Type Date Posted  Min Salary  Max Salary  \
0       Technology  Full-Time  2023-01-05     40000.0     60000.0
1        Analytics  Full-Time  2023-02-10     35000.0     50000.0
2       Networking  Full-Time  2023-03-15     45000.0     70000.0
3  Cloud Computing  Full-Time  2023-04-20     60000.0     90000.0
4         Security  Full-Time  2023-05-25     50000.0     80000.0

   Average Salary  Year
0         50000.0  2023
1         42500.0  2023
2         57500.0  2023
3         75000.0  2023
```

Salary increases with experience

Experience Level

Salary increases with experience

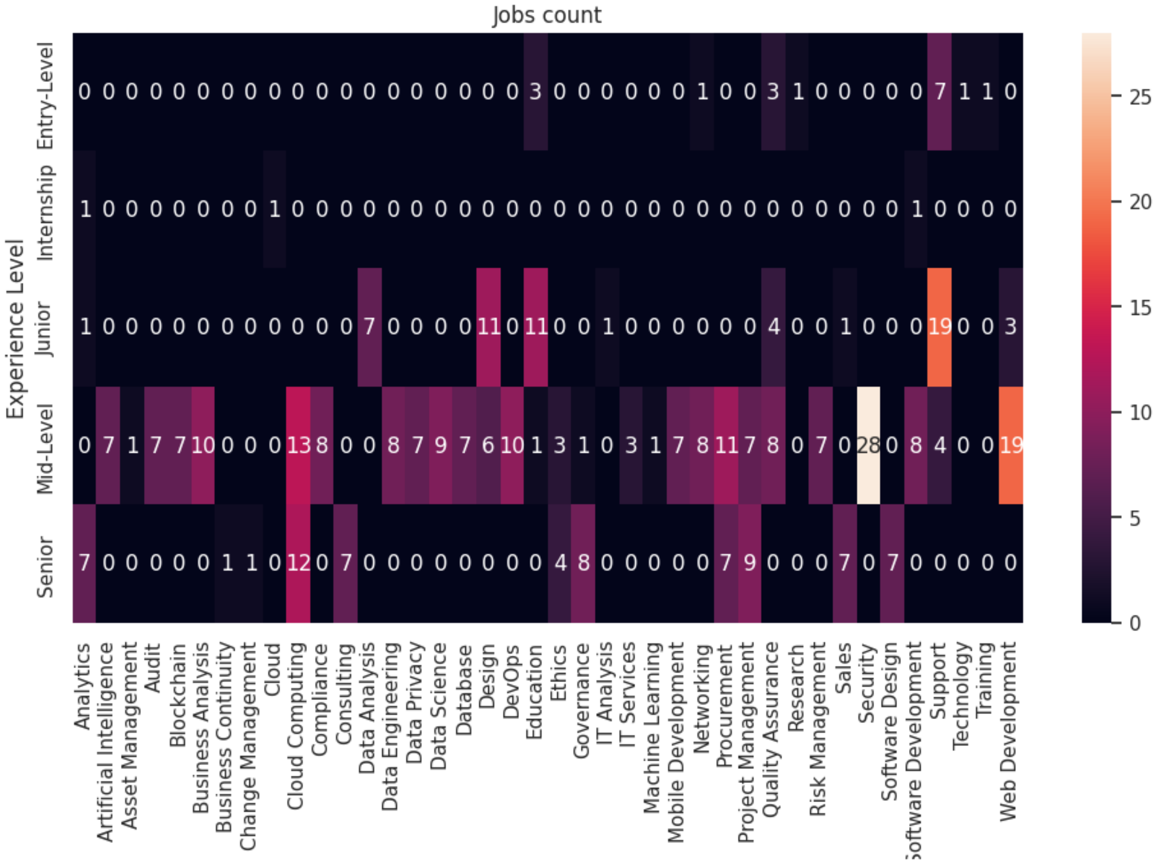


Different industries have different salaries

...

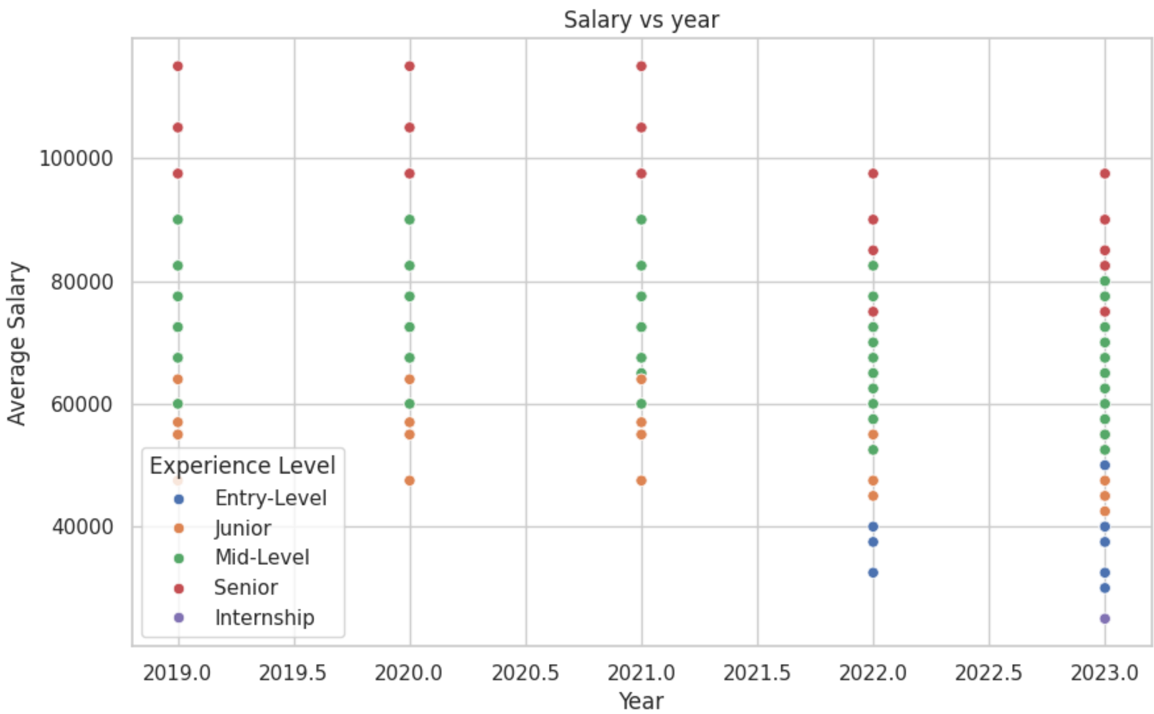
Industry

Different industries have different salaries



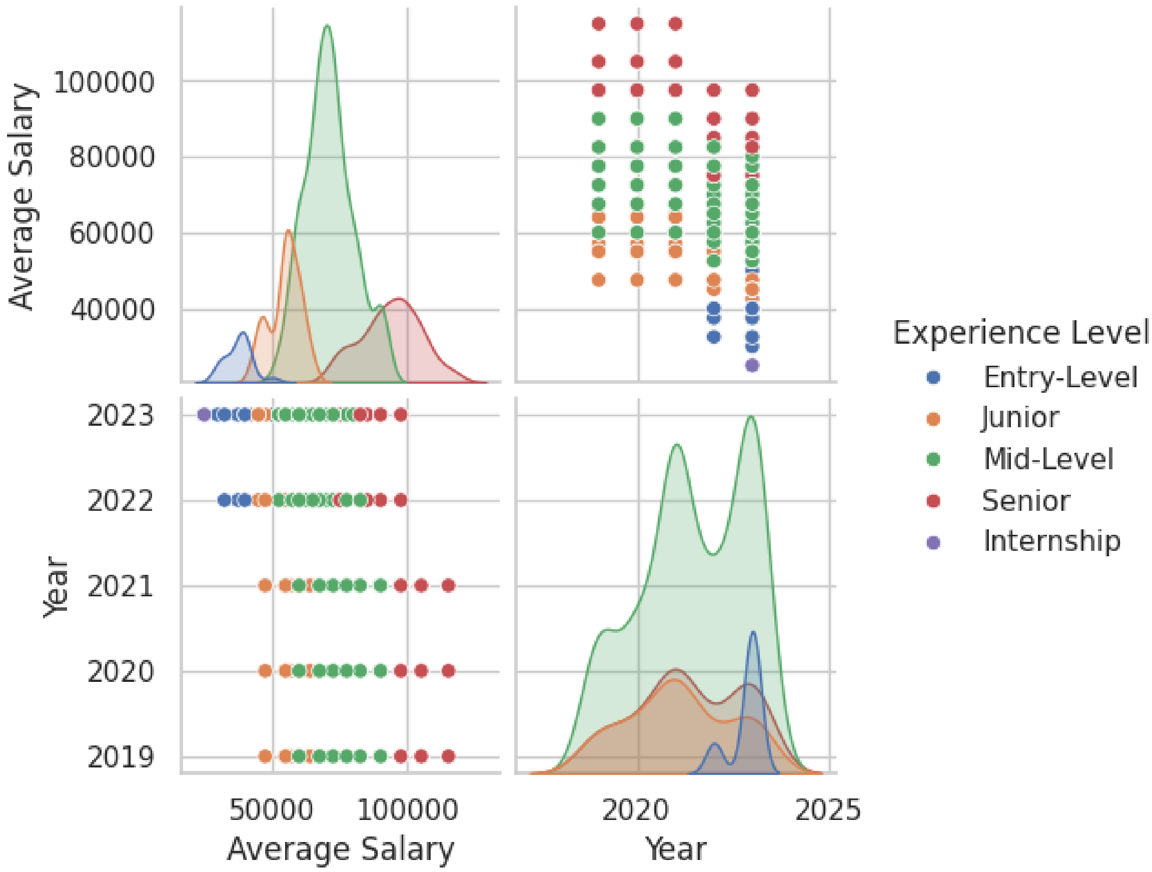
Industry

Most jobs are Mid and Senior



Salary grows with year

```
Salary grows with year
```




```
Relationships between variables
Seaborn helps visualize and analyze data
```

Висновок:
На цій лабораторній роботі я опрацював датасет вакансій та візуалізував дані за допомогою Seaborn. Було виявлено, що середня зарплата зростає разом із рівнем досвіду та різниться залежно від галузі. Найбільше вакансій пропонують для Mid та Senior рівнів, що підтверджує практичну користь аналізу даних для оцінки ринку праці.